sustainability

MDPI

Article

Moderating Impacts of Education Levels in the Energy–Growth–Environment Nexus

Busayo Victor Osuntuyi [1,2] and Hooi Hooi Lean [1,*]

[1] Economics Program, School of Social Sciences, Universiti Sains Malaysia, Gelugor 11800, Penang, Malaysia
[2] Department of Economics, Adekunle Ajasin University, Akungba-Akoko 342111, Nigeria
* Correspondence: hooilean@usm.my or learnmy@gmail.com

Abstract: The world's environment has deteriorated significantly over the years. Pollution's impact on the ecosystem is undeniably alarming. Many factors have been found in the literature to impact environmental pollution. However, there is a dearth of literature on the impacts of education levels on environmental pollution. This study, therefore, examines the effects of education levels and their moderating impacts on the energy–growth–environment nexus. Fundamentally, the study investigates the effects of economic growth, natural resources, and the marginal effects of energy consumption on environmental pollution at various levels of education in Africa from 1990 to 2017. The cross-sectional dependence test, unit root test, cointegration test, fixed effect estimation, Driscoll–Kraay standard errors, fully modified least ordinary least square estimator and dynamic ordinary least square estimator are employed for the analyses. The findings reveal that education increases environmental pollution and that the marginal impacts of energy consumption at various education levels adversely impact environmental pollution, implying that increased school enrollments exacerbate the adverse effects of energy consumption. The findings also show that economic growth, population, and trade openness degrade the environment, whereas natural resources promote environmental sustainability. We deduce several policy implications to improve environmental quality in Africa based on the findings.

Keywords: carbon emissions; economic growth; energy consumption; education; environmental pollution

check for updates

Citation: Osuntuyi, B.V.; Lean, H.H. Moderating Impacts of Education Levels in the Energy–Growth–Environment Nexus. *Sustainability* **2023**, *15*, 2659. https://doi.org/10.3390/su15032659

Academic Editors: José Alberto Fuinhas, Matheus Koengkan and Renato Filipe de Barros Santiago

Received: 10 November 2022
Revised: 17 January 2023
Accepted: 18 January 2023
Published: 1 February 2023

Copyright: © 2023 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (https://creativecommons.org/licenses/by/4.0/).

1. Introduction

Environmental pollution and climate change are the main challenges to sustainable global development and growth [1]. The environment of the world has deteriorated significantly over the past several decades. The environmental pollution impact on the planet's geography is profound, and environmental stakeholders are incredibly concerned. As a result, countries are being pushed to solve environmental issues while sustaining economic growth [2]. Environmental destruction in many countries is attributed mainly to human activities such as rapid industrialization, population growth, economic expansion, urbanization, natural resources depletion and overdependence on fossil fuel consumption [3–7]. Carbon dioxide (CO2) emissions are undeniably one of the primary contributors to environmental pollution [8–12]. As a result, various studies have employed CO2 emissions to measure the environmental impacts of human activities [7,13–16].

Understanding the mechanism of CO_2 emissions is crucial for energy decarbonization and climate change mitigation. Therefore, countries are devoting their resources to mitigating the effects of climate change and reducing the inherent vulnerabilities associated with frequent natural disasters. The countries are implementing new environmental strategies and regulations to alter their energy consumption patterns, rely more on renewable energy sources, and meet stricter environmental standards through advanced technologies in their manufacturing and consumption activities. Additionally, they are implementing various climate policies and coordinating several global efforts to mitigate the adverse

effects of global warming [17,18]. Some of these notable efforts are the Kyoto Protocol of 1997, the Paris Convention of 2015, and the Sustainable Development Goals (SDGs) of 2015. However, education is required for the potential positive impacts of government regulations, strategies, and policies on environmental quality. Without education, government regulatory measures, efforts, and policies will not be enough to promote environmentally friendly behaviors and accomplish environmental benefits [19].

Education can impact the environment by raising awareness and motivating people to protect it. It helps people reconsider their habits that are harmful to the environment, allowing them to utilize resources effectively and gain a deeper awareness of environmental challenges [20]. When it comes to environmental policy issues, educated people are more likely to back them. A country's reliance on a particular energy source is inextricably linked to its level of educational attainment [21]. Existing research, however, indicates that education might have two conflicting impacts on environmental quality. In societies with low levels of education, a boost in school enrollment would accelerate the consumption of non-renewable energy resources, causing environmental damage [19]. Increased knowledge in these countries increases the use of environmentally damaging products. On the other hand, when school enrollments reach certain levels, more education will boost eco-friendly technology and environmental knowledge [22]. Furthermore, more education attracts technology, and the level of education impacts how firms employ technologies [23,24].

The argument underlying education's ability to moderate the environmental impacts of energy consumption is straightforward. Theoretically, when education levels rise, the energy consumption structure might change dramatically by enabling consumers to substitute energy sources and comprehend detailed information about energy prices and usage [25]. Additionally, increased educational levels can result in more knowledgeable customers making more informed energy usage decisions, thereby moderating the environmental impact of energy consumption. However, the direction of such a possible impact needs to be empirically examined. Hence, it is crucial to investigate education's probable effects on environmental pollution through its moderating impacts on energy consumption. Therefore, the specific objective of this paper is to examine the impacts of education levels and their moderating effects on the energy–growth–environment nexus in Africa. Fundamentally, it seeks to ascertain the effects of economic growth, natural resources, and the marginal effects of energy consumption on environmental pollution at various levels of education. We test the hypothesis that economic growth, natural resources, energy consumption and education have no impacts on environmental pollution. We also test the hypothesis that education has no moderate effects on energy consumption.

This study contributes to the literature in three ways. First, we investigate the impacts of education levels on environmental pollution. Scarce attention is given to the impacts of education on environmental pollution, particularly in Africa. Such insufficient attention is somewhat surprising since the role of education in recognizing the causes of global climate change and its consequences on the environment cannot be denied. Most environmental literature ignores the effects of education levels on the environment. This study fills the gap by examining the effects of primary, secondary, and tertiary education levels on environmental pollution. Second, we examine the moderating impacts of energy consumption and education levels on environmental pollution. Most studies on education's influence on the environment assume only a direct effect. It is vital to emphasize that when the education variable takes a direct impact, the relationship between education and environmental pollution can only go one way. As a result, education's direct and moderating impacts cannot be studied separately. The current study is novel as it uses the moderating effects of education levels on environmental pollution. Third, we examine the contingency effects of education by calculating the marginal impacts of energy consumption on environmental pollution at various levels of education. Thus, we separately estimate education impacts at mean, minimum and maximum levels. From a policy standpoint, examining what happens at the mean, minimum, and maximum levels of primary, secondary, and tertiary education is critical. Ignoring such empirical analysis may result in incorrect policy recommendations.

There is a scarcity of evidence from research that has used marginal impacts to examine the link between education levels and environmental pollution.

To the best of our knowledge, this is the first study to examine the marginal impacts of energy consumption at different levels of education on environmental pollution in Africa. Africa is particularly vulnerable to the harmful effects of climate change. Hence, it is essential to investigate these implications. Africa's temperature is expected to climb faster than the world average, reaching 4–6 degrees Celsius this century [26]. Nevertheless, according to UNESCO 2019 reports, Africa has the greatest incidence of school exclusion in the world, with more than 20% of primary school-age children denied the right to education.

The rest of the paper is organized as follows: A review of the literature is presented in Section 2. The methodologies used in the study are described in Section 3. Section 4 presents and discusses the results. Section 5 concludes the paper.

2. Literature Review and Knowledge Gap

2.1. Literature Review

Extensive research has been conducted on the link between economic growth and the environment. The environmental Kuznets curve (EKC) hypothesis proposed by Grossman and Krueger [27] has been used in the environmental literature to identify pollution causes. The EKC hypothesis postulates that environmental pollution rises in the early stages of economic growth as economic output rises. This process will continue until a certain economic growth level is reached, at which point economic growth and environmental pollution will begin to diminish. However, empirical research is ambiguous regarding evidence to validate the EKC hypothesis. Some empirical research, for example, has offered evidence supporting the EKC hypothesis. For example, some empirical studies have provided proof for the EKC hypothesis [12,28–37]. In contrast, some studies have discovered no substantial evidence to support the EKC hypothesis [11,38–40].

Additionally, several empirical investigations have demonstrated that energy consumption increases pollution levels [8,41–45]. They documented that as energy usage rises, pollutant levels rise [46,47]. However, some studies reveal that renewable energy reduces pollution [3,4,48–50]. Another variable that influences environmental pollution is natural resources. Some scholars believe that natural resources significantly influence atmospheric pollutants and have reported mixed findings. For instance, Bekun et al. [51] and Sun et al. [52] found that natural resources have detrimental effects on environmental sustainability. In contrast, Balsalobre-Lorente et al. [53], Khan et al. [15] and Shittu et al. [54] found that natural resources boost environmental quality. Danish et al. [14] discovered that while an abundance of natural resources increases CO2 emissions in South Africa, it mitigates them in Russia. Correspondingly, Ahmad and Satrovic [55] found that the direct impacts of natural resources reduce energy and carbon efficiency while positively moderating the effects of fiscal decentralization and financial inclusion on environmental sustainability.

Education-related variables have recently been incorporated into the study. For example, Chankrajang and Muttarak [56] discovered that learning about eco-friendly conduct leads to pro-environmental decisions, aiding CO2 reduction and environmental protection. Furthermore, education is required to understand the worldwide repercussions of climate change and its negative consequences. Likewise, Balaguer and Cantavella [21] discovered that education enhances environmental quality in Australia. Other studies have confirmed Chankrajang and Muttarak's conclusions about the favorable effects of education on pro-environmental attitudes and environmental quality. Meyer [57], for example, noticed that educated individuals in Europe are more aware of the external consequences of their actions and, as a result, are more concerned with societal welfare.

Similarly, it was found that education improves the environment in Turkey [5] and the OECD [58], and APEC countries [59]. Nevertheless, Zafar et al. [60] discovered that education increases environmental deterioration to some extent, which does not enhance the quality of the environment. Likewise, Mahalik et al. [19] discovered that primary education increases environmental degradation in China, Brazil, India, and South Africa, whereas

secondary education enhances it. Maranzano et al. [61] discovered that the Educational EKC hypothesis holds for economies with significant income inequality and that the emissions–income elasticity seems to decrease when education is taken into account in OECD Countries. Conversely, Liu et al. [62] and Zhang et al. [63] found that education reduces environmental quality in Latin American and developing countries. Still, according to Boukhelkhal [64], education will increase environmental damage in the short run but decrease it in the long run in Algeria.

In examining the moderating effects of education, Katircioglu et al. [65] demonstrated that education's moderating function and direct influence on energy usage are harmful to the environment in Cyprus. Similarly, Osuntuyi and Lean [66] found that education worsens environmental deterioration. Its moderating influence, however, helps to mitigate energy consumption's negative effects on the environment in high- and upper-middle-income groups while amplifying them in low- and lower-middle-income groups. Similarly, Osuntuyi and Lean [67] discovered that the direct and moderating impacts of education exacerbate environmental pollution in Africa.

Education has also been employed as a proxy for human capital in other studies, with mixed results. Human capital, for example, has been demonstrated to lessen environmental degradation [23,31,68–71] without decreasing economic growth [72]. However, research by Zhang et al. [16] and Ahmed et al. [73] has established that human capital is harmful to the environment, refuting this assumption. On the other hand, Tang et al. [74] looked at the indirect and direct effects of human capital in 114 nations. They discovered that human capital has a significant impact on renewable energy use.

2.2. Knowledge Gap

The majority of studies on the impact of education on the environment [19,59,61–64] only consider direct impacts. Khan [23], Katircioglu et al. [65], Osuntuyi and Lean [66], and Osuntuyi and Lean [67] are notable exceptions. Khan [23] broadens the classic EKC model by including various levels of education variables and studying the moderating effects of economic growth and education levels on environmental quality. The study, however, did not consider the moderating roles of education and energy use. In contrast, Katircioglu et al. [65] only evaluated the moderating influence of higher education, whereas Osuntuyi and Lean [66] only examined the moderating impacts of primary education.

Additionally, there is a dearth of evidence from studies that have examined the relationship between education levels and environmental pollution by examining marginal impacts. Although Osuntuyi and Lean [67] evaluated the marginal effects of education, the study utilized only primary school education. Finally, researchers are divided on the environmental consequences of education, while its environment-moderating effects have received less attention. Variations in empirical methodology, data, time, or countries may have affected previous empirical findings.

3. Methodology

3.1. Model Specification

This study's theoretical framework is the EKC hypothesis of Grossman and Krueger [27]. Econometrically, the model is given as follows:

$$E = \beta_0 + \beta_1 Y_t + \beta_2 Y^2 + Z_t + \mu_t \quad (1)$$

where E_t is an indicator of environmental quality, Y_t is per capita GDP (as an economic growth measure), ${Y_t}^2$ is the square of per capita GDP, and μ_t is the normally distributed stochastic term. Z_t is a vector of additional variables that might influence environmental degradation. If $\beta_1 > 0$ and $\beta_2 < 0$, the link is inverted U-shaped, verifying the EKC hypothesis. However, if $\beta_1 < 0$ and $\beta_2 > 0$, it indicates the nexus is U-shaped. Thus, the hypothesis is not validated.

Following [27] model, the basic model for this study is specified as follows:

$$CO2_{it} = \delta_0 + \delta_1 GDP_{it} + \delta_2 GDP_{it}{}^2 + \delta_3 ENG_{it} + \delta_4 NRE_{it} + \delta_5 PRI_{it} + \delta_6 POP_{it} + \delta_7 TRD_{it} + \mu_{it} \quad (2)$$

$$CO2_{it} = \alpha_0 + \alpha_1 GDP_{it} + \alpha_2 GDP_{it}{}^2 + \alpha_3 ENG_{it} + \alpha_4 NRE_{it} + \alpha_5 SEC_{it} + \alpha_6 POP_{it} + \alpha_7 TRD_{it} + \mu_{it} \quad (3)$$

$$CO2_{it} = \chi_0 + \chi_1 GDP_{it} + \chi_2 GDP_{it}{}^2 + \chi_3 ENG_{it} + \chi_4 NRE_{it} + \chi_5 TER_{it} + \chi_6 POP_{it} + \chi_7 TRD_{it} + \mu_{it} \quad (4)$$

where *CO2* = carbon emissions (as an environmental pollution proxy), *GDP* = per capita real *GDP* (as an economic growth proxy), GDP^2 = square of *GDP*, *ENG* = energy consumption, *NRE* = natural resources, *PRI* = primary education, *SEC* = secondary education, *TER* = tertiary education, *POP* = population growth, *TRD* = trade openness, *i* = the country index and *t* = time index. Population and trade openness are added as control variables.

The literature has established that education determines how energy consumption choices affect the environment [25]. Therefore, to investigate the moderating roles of education, we first model the interaction between energy consumption and education variables as follows:

$$CO2_{it} = \gamma_0 + \gamma_1 GDP_{it} + \gamma_2 GDP_{it}{}^2 + \gamma_3 ENG_{it} + \gamma_4 NRE_{it} + \gamma_5 PRI_{it} + \gamma_6 (ENG * PRI)_{it} + \gamma_7 POP_{it} + \gamma_8 TRD_{it} + \mu_{it} \quad (5)$$

$$CO2_{it} = \eta_0 + \eta_1 GDP_{it} + \eta_2 GDP_{it}{}^2 + \eta_3 ENG_{it} + \eta_4 NRE_{it} + \eta_5 SEC_{it} + \eta_6 (ENG * SEC)_{it} + \eta_7 POP_{it} + \eta_8 TRD_{it} + \mu_{it} \quad (6)$$

$$CO2_{it} = \lambda_0 + \lambda_1 GDP_{it} + \lambda_2 GDP_{it}{}^2 + \lambda_3 ENG_{it} + \lambda_4 NRE_{it} + \lambda_5 TER_{it} + \lambda_6 (ENG * TER)_{it} + \lambda_7 POP_{it} + \lambda_8 TRD_{it} + \mu_{it} \quad (7)$$

Following that, we investigate energy consumption's marginal impacts on environmental pollution at various levels of education. We employ the partial derivatives of Equations (5)–(7) with respect to energy consumption to capture the marginal effects. The equations are specified below:

$$\frac{\partial CO2_{it}}{\partial ENG_{it}} = \gamma_3 + \gamma_6 PRI_{it} \quad (8)$$

$$\frac{\partial CO2_{it}}{\partial ENG_{it}} = \eta_3 + \eta_6 SEC_{it} \quad (9)$$

$$\frac{\partial CO2_{it}}{\partial ENG_{it}} = \lambda_3 + \lambda_6 TER_{it} \quad (10)$$

3.2. Data

The study uses annual time-series data from thirty-one African countries from 1990 to 2017. Data availability for a handful of selected nations over a long period determines the estimation timeframe. The countries are Algeria, Botswana, Benin Republic, Burkina Faso, Burundi, Cameroon, Côte d'Ivoire, Congo Republic, Egypt, Eswatini, Ethiopia, Ghana, Gambia, Guinea, Kenya, Lesotho, Madagascar, Mali, Mauritania, Mauritius, and Morocco. Others are Niger, Nigeria, Rwanda, South Africa, Senegal, Tanzania, Tunisia, Togo, Uganda and Zambia. Table 1 describes and provides the data sources for the study.

Table 1. Description of variables and sources of data.

Variables	Measurement	Sources
CO2	Thousand metric tons of CO2	WDI [75]
EFP	Measured in global hectares	EFP [76]
GDP	Constant 2010 US dollar	WDI [75]
ENG	British thermal units (Btu)	EIA [77]
NRE	A composite index of oil rents, natural gas rents, coal rents (hard and soft), mineral rents, and forest rents. Measured as a share of GDP	WDI [75]
PRI	Percentage of gross primary school enrollment	WDI [75]
SEC	Percentage of gross primary school enrollment	WDI [75]
TER	Percentage of gross tertiary school enrollment	WDI [75]
TRD	The sum of exports and imports of goods and services. Measured as a share of GDP	WDI [75]
POP	Midyear estimates	WDI [75]

Note: WDI = World Bank Development Indicator, EIA = Energy Information Administration.

Figure 1 reveals that all variables, including economic growth and its squared term, energy consumption, natural resources, education, population, and trade openness, as well as the moderating term between education variables and energy consumption, show either a positive or negative influence on environmental pollution in Africa.

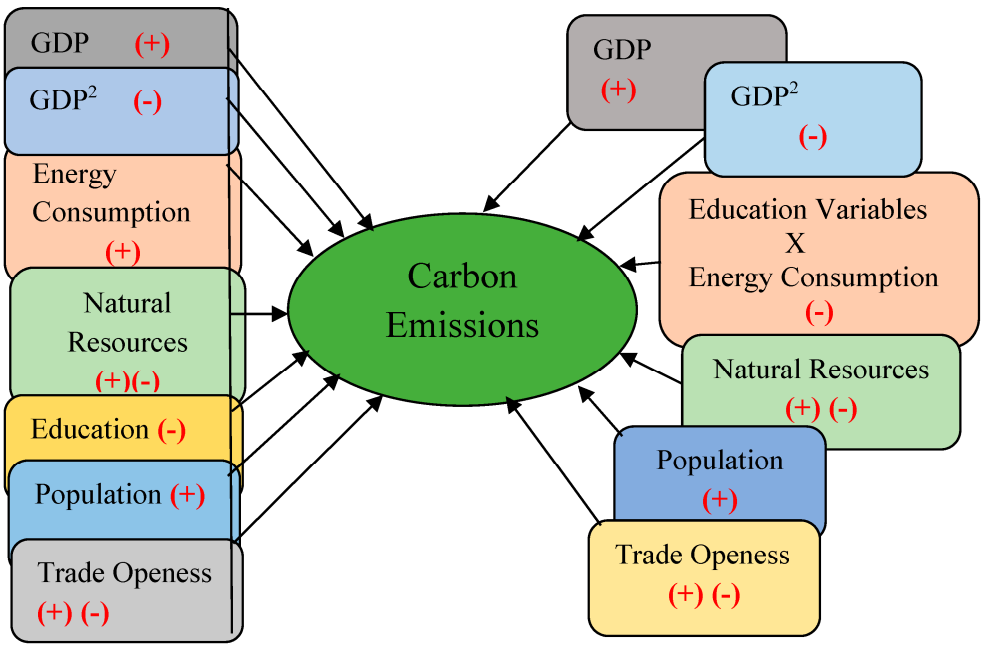

Figure 1. Conceptual framework of the variables.

3.3. Estimation Strategy

Descriptive and correlation analyses are the first steps in this study. Cross-sectional dependence (CSD), cointegration, and panel unit root tests are also investigated. We employ the Breusch and Pagan [78], Baltagi et al. [79] and Pesaran [80] tests. Second-generation stationarity tests are employed. The tests are the cross-sectionally augmented Dickey-Fuller test (CADF) and the cross-sectionally augmented IPS (CIPS) statistic [80,81]. The Westerlund [82] cointegration test is used to determine the cointegration among the variables. The test accounts for cross-section dependencies and non-strictly exogenous regressors.

We employ the variance inflation factor (VIF) to identify potential multicollinearity in all the models. The VIF values are less than 10, suggesting the absence of multicollinearity [83]. We also conducted the Breusch–Pagan/Cook–Weisberg and Wooldridge tests.

The results show the presence of heteroskedasticity and serial correlation in all the models. Thus, we employ the Driscoll and Kraay [84] (DK) standard errors (based on fixed effects estimation) for panel regressions because of heteroskedasticity, autocorrelation, and cross-sectional dependence in our models. The DK standard errors [85] are autocorrelation consistent and robust to heteroskedasticity and general forms of cross-sectional and temporal dependence [32,86,87]. The DK estimator works well with balanced and unbalanced panels and can deal with missing values [45,88,89]. Driscoll and Kraay [84] show, using large T asymptotics, that the basic non-parametric time-series covariance matrix estimator may be improved to be resilient to extremely general types of cross-sectional and temporal dependence. The adjustment of the standard error estimates ensures that the covariance matrix estimator is consistent, regardless of the cross-sectional dimension N. As a result, DK's technique avoids the shortcomings of other approaches, often inapplicable when the cross-sectional dimension N of a micro-econometric panel is high [85].

We employ the FMOLS and DOLS techniques for robustness checks. Pedroni [90] proposed the FMOLS heterogeneous panel cointegration method. The FMOLS is consistent and employs a non-parametric approach to endogeneity issues and autocorrelations [91,92]. It does not suffer significant distortions when endogeneity and heterogeneity exist [93]. It corrects serial correlation and simultaneous bias [94]. It also considers the issues related to the intercept and eliminates the missing variables biases and homogeneity restrictions [95]. On the other hand, the DOLS technique, developed by Kao and Chiang [96], is based on a parametric dynamic panel. The DOLS corrects endogeneity, simultaneity and serial correlation issues. It generates unbiased long-run estimates and supplements the static regression with leads, lags and regressors' contemporaneous values in the first difference [91,97,98]. It also has asymptotic efficiency and robustness in a small sample [97]. It also yields reliable estimates of explanatory variable coefficients in small samples [98]. Finally, we calculate the marginal impacts and compute the new standard errors based on Brambor [99]. The methodological framework of the study is show in Figure 2.

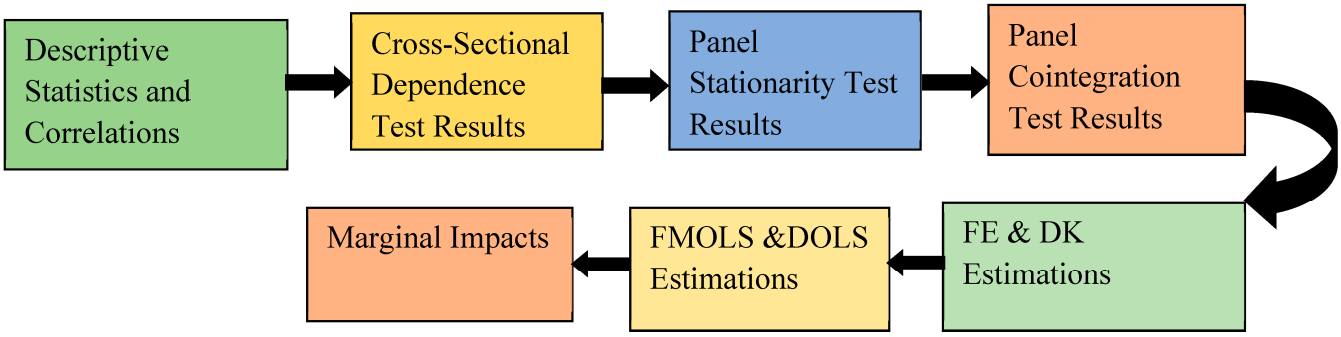

Figure 2. The methodological framework of the study.

4. Results and Discussion

4.1. Preliminary Analysis

Table 2 shows the descriptive statistics and correlation matrix of the variables used for estimations in this study. The standard deviation represents the dispersion from the mean value of each variable. Population variables have the highest standard deviation among the variables. The correlation coefficients in Table 2's lower panel reveal the correlation analysis between the dependent variable (CO2 emissions) and other variables. Except for the natural resources variable, the independent variables are positively associated with the dependent variable.

Table 2. Descriptive statistics and correlations.

Variable	CO2	EFP	GDP	ENG	NRE	PRI	SEC	TER	POP	TRD
Mean	26,776.04	35,502,934	1807.785	0.401	9.564	92.696	41.779	17.721	22,481,021	54.076
Minimum	160	1,014,855	164.337	0.003	0.001	21.708	5.221	0.332	822,423	4.104
Maximum	447,980	1.98×10^8	10,199.48	5.734	58.65	149.307	109.444	86.714	1.91×10^8	175.798
Std. Dev.	69,641.86	46,758,195	1836.281	1.009	9.038	25.376	25.74	21.63	29,072,716	30.126
CO2	1.000									
EFP	0.555	1.000								
GDP	0.555	0.155	1.000							
ENG	0.968	0.611	0.542	1.000						
NRE	−0.039	0.273	−0.526	−0.051	1.000					
PRI	0.251	0.003	0.39	0.205	−0.1	1.000				
SEC	0.632	0.252	0.771	0.587	−0.337	0.675	1.000			
TER	0.412	0.303	0.597	0.407	−0.194	0.502	0.737	1.000		
POP	0.565	0.573	−0.309	0.6	0.435	−0.133	−0.067	−0.095	1.000	
TRD	0.186	0.063	0.652	0.182	−0.441	0.391	0.531	0.487	−0.392	1.000

Notes: CO2 = carbon emissions, EFP = ecological footprint, GDP = real GDP per capita, ENG = energy consumption, NRE = natural resources, PRI = gross primary school enrollments, SEC = gross secondary school enrollments, TER = gross tertial education enrollments, POP = population, TRD = trade openness.

The CSD findings are shown in Table 3. The results of the tests indicate that the null hypothesis of no CSD is rejected at a 1% significant level for the panels. As a result, CSD exists among the panel countries, suggesting that any shock in one of the sample countries can spread to the others.

Table 3. Cross-sectional dependence test results.

Test	Breusch–Pagan LM	Pesaran Scaled LM	Bias-Corrected Scaled LM	Pesaran CD
	2008.3820 ***	50.6095 ***	50.0354 ***	0.5065 ***

Notes: *** denotes statistical significance at the 1% level.

Table 4 displays the panel unit root test results. The findings show that the null hypothesis of the presence of unit roots in the variables at different levels cannot be rejected, implying that the panel contains unit roots. Although some variables are stationary at levels using CADF, CIPS demonstrates that they are not. As a result, we conclude that such variables are not stationary at level.

Table 4. Panel stationarity test results.

Variables	CIPS		CADF	
	Level	Difference	Level	Difference
CO2	5.983	−15.210 ***	−1.529 *	−12.293 ***
EFP	4.978	−17.843 ***	−4.298 ***	−12.998 ***
GDP	5.421	−10.377 ***	−0.013	−8.134 ***
ENG	5.663	−14.885 ***	−1.711 **	−10.919 ***
NRE	−3.43553 **	−16.080 ***	−1.137	−12.634 ***
PRI	−1.589 *	−6.647 ***	−1.274	−5.208 ***
SEC	2.883	−5.006 ***	2.185	−5.007 ***
TER	3.141	−7.181 ***	2.453	−5.119 ***
POP	10.048	−20.557 ***	−13.65 ***	−11.030 ***
TRD	−2.13001 **	−15.453 ***	−0.010	−10.785 ***

Notes: ***, ** and * indicate statistical significance at 1%, 5% and 10%, respectively, and a rejection of the null hypothesis of unit root. CO2 = carbon emissions, EFP = ecological footprint, GDP = real GDP per capita, ENG = energy consumption, NRE = natural resources, PRI = gross primary school enrollments, SEC = gross secondary school enrollments, TER = gross tertial education enrollments, POP = population, TRD = trade openness.

We utilize the Westerlung cointegration test in Table 5. We conduct cointegration tests for different models based on the educational variable included in each of them. The test statistics confirm cointegration among the variables in each model at different significance levels. Therefore, we conclude that cointegration exists among the variables of the study. Since we have confirmed cointegration among the variables, we estimate the long-run relationships.

Table 5. Panel cointegration test results.

Model	Result
Model with Primary education	−2.854 ***
Model with Secondary education	−2.376 ***
Model With Tertiary education	−1.985 **

Notes: *** and ** indicates statistical significance at the 1% and 5% level of significance.

4.2. Estimation Results

Table 6 shows the estimated impact of economic growth, energy consumption, natural resources, and education on Africa's environmental pollution. Models 1–2 include the primary education variable, Models 3–4 contain the secondary education variable, and Models 5–6 have the tertiary education variable. Models 7–8, 9–10 and 11–12 incorporate interaction terms for the primary, secondary, and tertiary education variables. The results are generally similar.

Table 6. FE and DK estimations.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Variables	**FE**	**DK**	**FE**	**DK**	**FE**	**DK**	**FE**	**DK**	**FE**	**DK**	**FE**	**DK**
GDP	0.193 ***	0.193 ***	0.145 ***	0.145 **	0.252 ***	0.252 ***	0.191 ***	0.191 ***	0.180 ***	0.180 ***	0.253 ***	0.253 ***
	(0.050)	(0.054)	(0.054)	(0.066)	(0.059)	(0.048)	(0.053)	(0.058)	(0.062)	(0.049)	(0.064)	(0.049)
GDP^2	0.001	0.001	0.011	0.011	0.001	0.001	0.001	0.001	0.018	0.018	0.001	0.001
	(0.019)	(0.011)	(0.018)	(0.020)	(0.018)	(0.015)	(0.019)	(0.011)	(0.019)	(0.024)	(0.020)	(0.018)
ENG	0.461 ***	0.461 ***	0.349 ***	0.349 ***	0.273 ***	0.273 ***	0.461 ***	0.461 ***	0.343 ***	0.343 ***	0.274 ***	0.274 ***
	(0.029)	(0.054)	(0.031)	(0.048)	(0.032)	(0.040)	(0.029)	(0.054)	(0.031)	(0.046)	(0.035)	(0.047)
NRE	−0.052 ***	−0.052 ***	−0.027 *	−0.027 **	−0.020	−0.020	−0.053 ***	−0.053 ***	−0.025 *	−0.025 *	−0.020	−0.020
	(0.015)	(0.017)	(0.015)	(0.012)	(0.015)	(0.012)	(0.015)	(0.017)	(0.015)	(0.013)	(0.015)	(0.012)
PRI	0.128 ***	0.128 **					0.133 **	0.133 *				
	(0.046)	(0.055)					(0.060)	(0.067)				
SEC			0.148 ***	0.148 ***					0.120 ***	0.120 ***		
			(0.037)	(0.021)					(0.045)	(0.038)		
TER					0.093 ***	0.093 ***					0.093 ***	0.093 ***
					(0.022)	(0.013)					(0.025)	(0.015)
POP	0.978 ***	0.978 ***	0.996 ***	0.996 ***	1.095 ***	1.095 ***	0.979 ***	0.979 ***	1.009 ***	1.009 ***	1.095 ***	1.095 ***
	(0.065)	(0.090)	(0.081)	(0.050)	(0.073)	(0.037)	(0.065)	(0.090)	(0.082)	(0.050)	(0.074)	(0.035)
TRD	0.195 ***	0.195 ***	0.231 ***	0.231 ***	0.188 ***	0.188 ***	0.195 ***	0.195 ***	0.234 ***	0.234 ***	0.189 ***	0.189 ***
	(0.032)	(0.048)	(0.032)	(0.027)	(0.033)	(0.031)	(0.032)	(0.047)	(0.032)	(0.027)	(0.034)	(0.032)
ENG×PRI							0.003	0.003				
							(0.027)	(0.012)				
ENG×SEC									−0.016	−0.016		
									(0.014)	(0.012)		
ENG×TER											−0.001	−0.001
											(0.009)	(0.008)
Constant	−8.881 ***	−8.881 ***	−9.190 ***	−9.190 ***	−11.335 ***	−11.335 ***	−8.895 ***	−8.895 ***	−9.583 ***	−9.583 ***	−11.348 ***	−11.348 ***
	(1.038)	(1.663)	(1.295)	(0.990)	(1.304)	(0.829)	(1.044)	(1.646)	(1.341)	(0.823)	(1.323)	(0.753)
Observations	810	810	566	566	568	568	810	810	566	566	568	568
No of Countries	31		31		31		31		31		31	
Mean VIF	6.40		7.78		7.05		6.20		7.39		7.93	
Heterosk-edasticity	37.48 [000]		27.88 [000]		39.20 [000]		37.13 [000]		28.74 [000]		35.60 [000]	
Serial Correlation	66.846 [000]		52.398 [000]		65.466 [000]		60.555 [000]		52.122 [000]		65.454 [000]	

Notes: ***, ** and * indicate statistical significance at 1%, 5% and 10%, respectively. Standard errors in parenthesis. Probability values are in the bracket.

The results show that economic growth is positive and statistically significant in all our estimations. Our findings are consistent with those of previous studies. The findings indicate that Africa may have neglected environmental preservation to achieve economic growth. The present economic growth may be unsustainable because of the African economy's dependence on fossil fuels. According to the findings, a slowdown in economic development might positively impact the environment. On the contrary, this strategy is not viable since policy intervention must consider harmful effects on environmental quality via economic growth. It is probable that due to the continent's sustained economic growth

trend, environmental pollution may become a problem. This argument becomes apparent when analyzing the effects of the squared term of economic growth. The squared terms of economic growth coefficients are insignificant in our estimations. These findings imply that the EKC hypothesis is not feasible in Africa. They also support the view that the EKC does not appear by chance and underline the need for Africa-tailored policies that counteract the negative environmental impacts of economic growth with education.

The positive nexus between CO2 emissions and the energy consumption coefficient is positive and statistically significant, suggesting that energy consumption significantly aggravates environmental pollution in Africa. These findings are in line with those of Ehigiamusoe [4] and Zafar [60]; however, they contradict those of others who have discovered that using renewable energy reduces CO2 emissions [59,74]. This result can be ascribed to Africa's growing reliance on fossil fuel energy. In Africa, fossil fuel consumption accounts for more than 90% of overall energy consumption. Increased fossil fuel use results in higher CO2 release [100]. Furthermore, Africa's rapid industrialization and transportation network growth have boosted energy consumption, resulting in negative environmental externalities such as increasing CO2 emissions. As African economies rise, so does the need for fossil fuels for commercial, residential, and industrial usage, resulting in rising greenhouse gas (GHG) emissions.

The negative relationship between natural resources and CO2 emissions illustrates that increasing natural resource usage decreases pollution in Africa. These findings agree with those of Danish et al. [14], Balsalobre-Lorente et al. [101], Khan [15], Zhang et al. [16] and many others but differ from the findings of [102]. These findings imply that natural resource utilization helps improve Africa's environmental quality. Additionally, the findings confirm that countries with abundant natural resources have a higher environmental quality standard than those with a scarcity of natural resources [16].

Surprisingly, primary, secondary and tertiary education levels positively affect CO2 emissions, implying that they adversely influence environmental pollution in Africa. These findings corroborate those of Ahmed et al. [73], Katircioglu et al. [65] and Zafar et al. [60], and Zafar et al. [103]. They discovered that education contributes to environmental pollution. However, this discovery contradicts previous research indicating that education lowers environmental pollution [5,21,59,72]. According to the findings, a higher level of education facilitates access to energy-intensive technology. These findings could be explained by the fact that education programs in Africa lack specialized content on environmental sustainability. Without energy-saving training and targeted environmental awareness initiatives, education will likely foster a more resource-intensive, affluent lifestyle, contributing to environmental pollution [73]. These findings show that education cannot help lessen environmental pollution without an environmentally sustainable curriculum. A comprehensive set of environmental rules is essential to derive any value from education. Otherwise, education will increase people's purchasing power, energy usage, and usage of unsustainable natural resources, resulting in environmental pollution. Incorporating environmental content into education, raising media awareness, and providing energy efficiency training are all possible policy options for boosting the environmental benefits of education [103].

The findings also show that the interaction term coefficients between energy consumption and education variables are not statistically significant. However, just because the coefficients of the interaction terms are insignificant does not imply that the variables do not interact [74]. The reason is that the interaction terms are not interpreted as unconditional effects. Thus, we need to calculate the marginal impacts [99]. Section 4.4 examines the marginal impacts of energy consumption at different educational levels.

The population coefficient shows significantly positive relationships with CO2 emissions in all our models, suggesting that the population causes serious environmental pollution in Africa. The findings concur with those of Hanif et al. [48], Ohlan [104] and Wang et al. [105]. With the end of colonialism in Africa came a striking expansion of social services across the continent, notably in healthcare and education. Due to this increase, infant mortality significantly decreased, and population growth quickly increased [106]. Africa is the fastest-growing continent, with the highest population growth rate. The population of Sub-Saharan Africa is anticipated to double by 2050 [107]. Similarly, the empirical findings also indicate that trade openness increases environmental pollution. The findings align with Ali et al. [13], Abid [38] and Pata and Caglar [70]. These findings might be because, during an earlier stage of development, the primary objective of African policymakers was to achieve growth, even at the expense of the environment. As a result, low-cost, polluting technologies were introduced into African nations to promote output, and the technical impact of trade openness deteriorated environmental quality in the process [108].

4.3. Robustness Checks

To determine how sensitive the results are to various estimation strategies and methods, we performed a number of robustness checks. First, we used the FMOLS and DOLS methodologies to assess the robustness of our estimation results. The results in Table 7 are similar to those in Table 6. The coefficients have comparable signs, sizes, and significance. The findings confirm that our estimations are robust. Second, the robustness checks involve using ecological footprints as an alternative proxy for environmental pollution. The results are shown in Table 8. The results are similar to our earlier results in Table 6 except for some variables. The findings show that energy consumption and natural resource use increase environmental pollution, while trade openness exerts negative relationships with environmental pollution. Education variables have mixed impacts. Primary education is significantly positive, implying that primary education contributes to environmental degradation. On the other hand, the secondary education variable is significantly negative, indicating that secondary education reduces environmental degradation. In contrast, tertiary education has no significant impact on environmental pollution in Africa. Moreover, the findings indicate that the moderating role of education and energy consumption exacerbates environmental pollution.

Table 7. FMOLS and DOLS estimations.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Variables	**FMOLS**	**DOLS**	**FMOLS**	**DOLS**	**FMOLS**	**DOLS**	**FMOLS**	**DOLS**	**FMOLS**	**DOLS**	**FMOLS**	**DOLS**
GDP	0.176 **	0.275 ***	−0.004	−0.053	0.250 **	0.190	0.186 **	0.300 ***	0.047	0.511 ***	0.308 ***	0.165
	(0.077)	(0.074)	(0.084)	(0.094)	(0.098)	(0.138)	(0.080)	(0.079)	(0.098)	(0.196)	(0.109)	(0.150)
GDP^2	0.016	−0.033	0.015	−0.075*	0.007	−0.048	0.019	−0.035	0.023	−0.263 **	0.029	−0.011
	(0.030)	(0.035)	(0.029)	(0.040)	(0.030)	(0.060)	(0.029)	(0.036)	(0.030)	(0.109)	(0.034)	(0.072)
ENG	0.471 ***	0.316 ***	0.410 ***	0.237 ***	0.240 ***	0.283 ***	0.462 ***	0.311 ***	0.396	0.168 ***	0.265 ***	0.249 ***
	(0.044)	(0.040)	(0.047)	(0.044)	(0.055)	(0.073)	(0.043)	(0.040)	(0.048)	(0.056)	(0.060)	(0.079)
NRE	−0.063 ***	−0.038 **	−0.004	−0.025	−0.019	−0.001	−0.060 ***	−0.033 *	−0.003	−0.030	−0.021	−0.004
	(0.022)	(0.019)	(0.021)	(0.020)	(0.023)	(0.025)	(0.022)	(0.020)	(0.021)	(0.025)	(0.023)	(0.026)
PRI	0.152 **	0.152 **					0.160*	0.054				
	(0.070)	(0.070)					(0.089)	(0.091)				
SEC			0.110 *	0.196 ***					0.073 ***	0.022		
			(0.062)	(0.050)					(0.073)	(0.080)		
TER					0.106 ***	0.106 **					0.082 *	0.092 *
					(0.038)	(0.045)					(0.042)	(0.048)
POP	0.991 ***	1.116 ***	1.115 ***	1.464 ***	1.118 ***	1.054 ***	0.990 ***	1.130 ***	1.136 ***	1.120 *	1.141 ***	1.130 ***
	(0.103)	(0.094)	(0.137)	(0.130)	(0.128)	(0.186)	(0.102)	(0.097)	(0.137)	(0.616)	(0.132)	(0.204)
TRD	0.231 ***	0.116 ***	0.233 ***	0.080 *	0.237 ***	0.116 *	0.227 ***	0.122 ***	0.232 ***	0.061	0.258 ***	0.106 *
	(0.049)	(0.041)	(0.052)	(0.044)	(0.054)	(0.059)	(0.048)	(0.042)	(0.052)	(0.060)	(0.056)	(0.061)
ENG×PRI							0.012	0.054				
							(0.040)	(0.042)				
ENG×SEC									0.019	0.053		
									(0.021)	(0.043)		
ENG×TER											0.018	0.001
											(0.015)	(0.018)

Notes: ***, ** and * indicate statistical significance at 1%, 5% and 10%, respectively. Standard errors in parenthesis.

Table 8. FE and DK estimations using ecological footprints as a proxy for a robustness check.

	(2)	(3)	(7)	(8)	(2)	(3)	(2)	(3)	(2)	(3)	(2)	(3)
Variables	**FE**	**DK**	**FE**	**DK**	**FE**	**DK**	**FE**	**DK**	**FEM**	**DK**	**FE**	**DK**
GDP	0.307 ***	0.307 ***	0.377 ***	0.377 ***	0.385 ***	0.385 ***	0.259 ***	0.259 ***	0.343 ***	0.343 ***	0.254 ***	0.254 ***
	(0.028)	(0.026)	(0.033)	(0.031)	(0.037)	(0.053)	(0.029)	(0.042)	(0.038)	(0.033)	(0.038)	(0.033)
GDP^2	0.129 ***	0.129 ***	0.099 ***	0.099 ***	0.111 ***	0.111 ***	0.130 ***	0.130 ***	0.092 ***	0.092 ***	0.066 ***	0.066 ***
	(0.011)	(0.004)	(0.011)	(0.007)	(0.012)	(0.009)	(0.011)	(0.005)	(0.012)	(0.009)	(0.012)	(0.012)
ENG	0.036 **	0.036**	0.025	0.025	0.001	0.001	0.031 *	0.031 ***	0.030	0.030 **	−0.072 ***	−0.072 ***
	(0.016)	(0.014)	(0.019)	(0.016)	(0.020)	(0.027)	(0.016)	(0.010)	(0.019)	(0.014)	(0.021)	(0.025)
NRE	0.057 ***	0.057 ***	0.056 ***	0.056 ***	0.057 ***	0.057 ***	0.051 ***	0.051 ***	0.055 ***	0.055 ***	0.058 ***	0.058 ***
	(0.008)	(0.011)	(0.009)	(0.014)	(0.009)	(0.017)	(0.008)	(0.010)	(0.009)	(0.014)	(0.009)	(0.018)
PRI	0.120 ***	0.120 ***					0.234 ***	0.234 **				
	(0.025)	(0.041)					(0.033)	(0.095)				
SEC			−0.096 ***	−0.096 **					−0.070**	−0.070 *		
			(0.023)	(0.037)					(0.027)	(0.038)		
TER					−0.011	−0.011					0.044 ***	0.044 **
					(0.014)	(0.018)					(0.015)	(0.020)
POP	0.748 ***	0.748 ***	1.042 ***	1.042 ***	0.928 ***	0.928 ***	0.756 ***	0.756 ***	1.029 ***	1.029 ***	0.905 ***	0.905 ***
	(0.036)	(0.049)	(0.050)	(0.033)	(0.046)	(0.033)	(0.035)	(0.041)	(0.050)	(0.033)	(0.043)	(0.030)
TRD	−0.073 ***	−0.073 ***	−0.077 ***	−0.077 ***	−0.106 ***	−0.106 ***	−0.080 ***	−0.080 ***	−0.079 ***	−0.079 ***	−0.143 ***	−0.143 ***
	(0.017)	(0.009)	(0.020)	(0.020)	(0.021)	(0.021)	(0.017)	(0.007)	(0.020)	(0.022)	(0.020)	(0.024)
ENG×PRI							0.078 ***	0.078 **				
							(0.014)	(0.037)				
ENG×SEC									0.015 *	0.015 ***		
									(0.009)	(0.005)		
ENG×TER											0.046 ***	0.046 ***
											(0.005)	(0.004)
Constant	1.986 ***	1.986 **	−2.322 ***	−2.322 ***	−0.838	−0.838	1.679 ***	1.679 ***	−1.945 **	−1.945 ***	0.300	0.300
	(0.572)	(0.789)	(0.790)	(0.645)	(0.822)	(0.798)	(0.565)	(0.562)	(0.817)	(0.667)	(0.777)	(0.632)
Observations	810	810	566	566	568	568	810	810	566	566	568	568
Number of Countries	31		31		31		31		31		31	
Mean VIF	7.26		8.85		7.99		6.20		7.39		7.93	
Heterosk-edasticity	39.37 [0.000]		22.26 [0.000]		23.22 [0.000]		34.25 [000]		15.02 [000]		16.09 [0.000]	
Serial Correlation	21.557 [0.000]		6.050 [0.0201]		9.343 [0.0048]		21.523 [000]		5.931 [0.0213]		9.810 [0.0048]	

Notes: ***, ** and * indicate statistical significance at 1%, 5% and 10%, respectively. Standard errors in parenthesis. Probability values are in the brackets.

4.4. Marginal Impacts

Table 9 shows the results of the marginal impacts. Following that, we discuss the marginal impact of energy usage on environmental pollution at the minimum, mean and maximum levels of education. Our findings show that the marginal impacts of energy consumption at various levels of education (primary, secondary and tertiary) positively impact environmental pollution. The results support Osuntuyi and Lean [67], who revealed that education had direct and moderating effects on environmental pollution. According to the findings, higher school enrollments intensify the negative consequences of energy consumption in Africa. Again, these findings can be attributed to the absence of environmental awareness content in African countries' curricula. When the magnitudes of the impacts are compared, primary schooling has the most negative marginal impacts on energy consumption. The amplitude of the impacts decreases with secondary and tertiary education variables. These findings show that low-educated individuals lack an understanding of environmental quality and consume more non-renewable energy, contributing to environmental damage. In comparison, those with better education and income would purchase energy-saving technology that is less harmful to the natural environment [19].

Table 9. Marginal impacts of energy consumption (at different levels of education).

		Minimum	**Mean**	**Maximum**
ENG×PRI	$\partial CO2_{it}/\partial ENG_{it}$	0.471 ***	0.476 ***	0.478 ***
		(0.077)	(0.091)	(0.097)
ENG×SEC	$\partial CO2_{it}/\partial ENG_{it}$	0.316 ***	0.287 ***	0.268 ***
		(0.034)	(0.032)	(0.039)
ENG×TER	$\partial CO2_{it}/\partial ENG_{it}$	0.274 ***	0.273 ***	0.272 ***
		(0.053)	(0.039)	(0.037)

Note: *** indicates statistical significance at a 1% level. Figures in parenthesis are standard errors calculated based on Brambor [99]. The marginal impacts are based on the results of DK estimation.

5. Conclusions and Policy Implications

The impact of pollution on the environment is undoubtedly alarming. Over time, the world's environment has degraded substantially. Several variables have been identified in

the literature as impacting environmental pollution. However, the literature on the impacts of education levels on environmental pollution is limited. As a result, from 1990 to 2017, this study investigated the moderating impacts of education levels in the energy–growth–environment nexus in Africa. We employed fixed effect estimation and Driscoll–Kraay standard errors and computed the marginal impacts of education. For a robustness check, we used the FMOLS estimator and DOLS estimator. We also utilized the ecological footprint as an additional proxy for environmental pollution. The paper's empirical findings yield several intriguing inferences with significant policy implications.

The findings indicate that pollution rises as the economy grows. However, the findings show that the squared terms of economic growth coefficients are insignificant, suggesting that the EKC is invalid. The results also demonstrate that energy consumption significantly increases environmental pollution in Africa. On the other hand, natural resources were found to reduce environmental pollution. Education levels at the primary, secondary and tertiary levels have been shown to have adverse effects on environmental pollution. Additionally, the findings reveal no statistical significance for the interaction term coefficients between energy consumption and the education variables. The results also divulge that the marginal effects of energy consumption at different levels of education contribute to environmental damage. According to the findings, higher school enrollments exacerbate the negative impacts of energy consumption in Africa. In all of our models, the population coefficient has a significantly positive relationship with CO2 emissions, suggesting that the population causes serious environmental pollution. Similarly, the empirical findings show that trade openness contributes to environmental damage in Africa.

Based on a shred of empirical evidence, this study highlights several policy implications for environmental sustainability. The invalidity of the EKC hypothesis demonstrates that economic growth is not a remedy for Africa's environmental pollution. Thus, effective measures are required to reduce environmental pollution significantly and promptly. The massive use of fossil fuels to generate economic growth has aggravated African environmental pollution. Therefore, Africa should decrease its dependency on fossil fuels while increasing its use of renewable energy sources.

Furthermore, governments in Africa must develop legislation to educate the populace about natural resource exploitation to prevent deforestation and land degradation, which will mitigate sustainability problems. Increasing awareness and leveraging governmental regulatory pressures may be a way to address environmental sustainability issues. Additionally, decision-makers should implement measures to balance the demand and supply of resources that will eventually contribute to preserving the environment's quality. The negative consequences of education on Africa's environment show that education alone will not result in a more ecologically conscious attitude or better environmental quality. School curricula must be transformed to foster environmental knowledge, competence, and mindset to combat environmental pollution. Hence, environmental education must be incorporated into the African education curriculum, necessitating interactive teaching and learning approaches that encourage and empower people to alter their environmental behavior and take action for sustainable development. Incorporating environmental sustainability knowledge and practices into African countries' education curricula can promote the sustainable use of energy and reduce environmental concerns in Africa.

Additionally, education is a crucial tool for implementing sustainable development in Africa. It offers an essential framework for integrating apparent social, economic, and environmental conflicts into a coherent idea and the goal of sustainable well-being for all. This extends beyond the fact that education is listed as a single sustainable development goal, necessitating a more profound comprehension of education's function as a cross-cutting implementation strategy to boost accomplishments across many other goals in Africa. A more comprehensive knowledge of environmental education creates a stronger mechanism for promoting environmental sustainability. Therefore, achieving a sustainable development agenda in Africa depends on people having appropriate envi-

ronmental knowledge and adopting proactive attitudes toward resolving environmental issues throughout their lifetimes.

The findings that population growth promotes environmental deterioration highlight the importance of increasing environmental awareness programs, encouraging people to adopt eco-friendly lifestyles, and rigorously monitoring the impacts of population growth on environmental sustainability. Furthermore, African authorities should use international trade to preserve environmental quality. Because small-scale industry players may be unable to produce endogenous clean production methods, trade openness might be leveraged to import cleaner technologies for those firms. With this approach, players at different industry levels will have enough time to establish their production methods and capitalize on the benefits of imported technology during that period.

In addition, this study's outcomes show that some variables' environmental impacts vary depending on the environmental indicator used. This could be because CO2 emissions only account for a portion of environmental damage, whereas ecological footprint provides a more comprehensive environmental sustainability assessment. As a result, Africa should implement comprehensive environmental policies that consider not just carbon emissions but also ecological footprint components such as built-up land, cropland, carbon absorption land, fishing grounds, forest area and grazing land.

This study, like so many others, has limitations. One potential drawback of the study is that we employ data from several institutions, which may have measurement errors. Furthermore, the study's empirical approach uses the DK, FMOLS, and DOLS estimators. Different panel data techniques may provide different outcomes. As a result, future studies using data from an extended period and new methods could be conducted to verify the validity of the findings of this study. Moreover, because some data were unavailable for an extended period, no country-specific analysis was carried out. Given this constraint, future research should concentrate on time-series analysis at the national level. Exploring the relationship between these variables within each country will be necessary to understand the relationship fully and will be critical for directing sustainable development policy. Additionally, total energy was used in the study. Future research could look at the relationship using disaggregated data. Finally, future studies should use the per capita version of carbon emissions and energy consumption rather than the absolute figures used in the current study.

Author Contributions: B.V.O.: conceptualization, data curation, formal analysis, visualization, writing—original draft, writing—review and editing. H.H.L.: conceptualization, investigation, resources, supervision, writing—original draft, writing—review and editing. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: Data are available upon request.

Conflicts of Interest: The authors declare no conflict of interest.

References

1. Verbič, M.; Satrovic, E.; Mujtaba, A. Assessing the Driving Factors of Carbon Dioxide and Total Greenhouse Gas Emissions to Maintain Environmental Sustainability in Southeastern Europe. *Int. J. Environ. Res.* **2022**, *16*, 105. [CrossRef]
2. Chomać-Pierzecka, E.; Kokiel, A.; Rogozińska-Mitrut, J.; Sobczak, A.; Soboń, D.; Stasiak, J. Analysis and Evaluation of the Photovoltaic Market in Poland and the Baltic States. *Energies* **2022**, *15*, 669. [CrossRef]
3. Destek, M.A.; Aslan, A. Disaggregated renewable energy consumption and environmental pollution nexus in G-7 countries. *Renew. Energy* **2020**, *151*, 1298–1306. [CrossRef]
4. Ehigiamusoe, K.U. The drivers of environmental degradation in ASEAN + China: Do financial development and urbanization have any moderating effect? *Singap. Econ. Rev.* **2020**, 1–44. [CrossRef]

5. Eyuboglu, K.; Uzar, U. A new perspective to environmental degradation: The linkages between higher education and CO_2 emissions. *Environ. Sci. Pollut. Res.* **2021**, *28*, 482–493. [CrossRef] [PubMed]
6. Mohsin, M.; Kamran, H.W.; Atif Nawaz, M.; Sajjad Hussain, M.; Dahri, A.S. Assessing the impact of transition from nonrenewable to renewable energy consumption on economic growth-environmental nexus from developing Asian economies. *J. Environ. Manag.* **2021**, *284*, 111999. [CrossRef]
7. Usman, O.; Alola, A.A.; Sarkodie, S.A. Assessment of the role of renewable energy consumption and trade policy on environmental degradation using innovation accounting: Evidence from the US. *Renew. Energy* **2020**, *150*, 266–277. [CrossRef]
8. Abokyi, E.; Appiah-Konadu, P.; Abokyi, F.; Oteng-Abayie, E.F. Industrial growth and emissions of CO_2 in Ghana: The role of financial development and fossil fuel consumption. *Energy Rep.* **2019**, *5*, 1339–1353. [CrossRef]
9. Castellanos, P.M.A.; Queiruga-Dios, A.; Encinas, A.H.; Acosta, L.C. Environmental education in environmental engineering: Analysis of the situation in Colombia and Latin America. *Sustainability* **2021**, *12*, 7239. [CrossRef]
10. Ehigiamusoe, K.U.; Lean, H.H.; Smyth, R. The moderating role of energy consumption in the carbon emissions-income nexus in middle-income countries. *Appl. Energy* **2020**, *261*, 114215. [CrossRef]
11. Ehigiamusoe, K.U.; Lean, H.H. Effects of energy consumption, economic growth, and financial development on carbon emissions: Evidence from heterogeneous income groups. *Environ. Sci. Pollut. Res.* **2019**, *26*, 22611–22624. [CrossRef]
12. Sirag, A.; Matemilola, B.T.; Law, S.H.; Bany-Ariffin, A.N.; Tunde Matemilola, B.; Bany-Ariffin, A.N. Does environmental Kuznets curve hypothesis exist? Evidence from dynamic panel threshold. *J. Environ. Econ. Policy* **2018**, *7*, 145–165. [CrossRef]
13. Ali, H.S.; Zeqiraj, V.; Lin, W.L.; Law, S.H.; Yusop, Z.; Bare, U.A.A.; Chin, L. Does quality institutions promote environmental quality? *Environ. Sci. Pollut. Res.* **2019**, *26*, 10446–10456. [CrossRef]
14. Danish; Baloch, M.A.; Mahmood, N.; Zhang, J.W. Effect of natural resources, renewable energy and economic development on CO_2 emissions in BRICS countries. *Sci. Total Environ.* **2019**, *678*, 632–638. [CrossRef] [PubMed]
15. Khan, I.; Hou, F.; Le, H.P. The impact of natural resources, energy consumption, and population growth on environmental quality: Fresh evidence from the United States of America. *Sci. Total Environ.* **2021**, *754*, 142222. [CrossRef] [PubMed]
16. Zhang, L.; Godil, D.I.; Bibi, M.; Khan, M.K.; Sarwat, S.; Anser, M.K. Caring for the environment: How human capital, natural resources, and economic growth interact with environmental degradation in Pakistan? A dynamic ARDL approach. *Sci. Total Environ.* **2021**, *774*, 145553. [CrossRef]
17. Amuakwa-Mensah, F.; Adom, P.K. Quality of institution and the FEG (forest, energy intensity, and globalization) -environment relationships in sub-Saharan Africa. *Environ. Sci. Pollut. Res.* **2017**, *24*, 17455–17473. [CrossRef]
18. Chomać-Pierzecka, E.; Sobczak, A.; Soboń, D. Wind Energy Market in Poland in the Background of the Baltic Sea Bordering Countries in the Era of the COVID-19 Pandemic. *Energies* **2022**, *15*, 2470. [CrossRef]
19. Mahalik, M.K.; Mallick, H.; Padhan, H. Do educational levels influence the environmental quality? The role of renewable and non-renewable energy demand in selected BRICS countries with a new policy perspective. *Renew. Energy* **2021**, *164*, 419–432. [CrossRef]
20. Ulucak, R.; Bilgili, F. A reinvestigation of EKC model by ecological footprint measurement for high, middle and low income countries. *J. Clean. Prod.* **2018**, *188*, 144–157. [CrossRef]
21. Balaguer, J.; Cantavella, M. The role of education in the Environmental Kuznets curve. Evidence from Australian data. *Energy Econ* **2018**, *70*, 289–296. [CrossRef]
22. Williamson, C. Emission, education, and politics: An empirical study of the carbon dioxide and methane environmental Kuznets curve. *Park Place Econ.* **2017**, *25*, 20–33.
23. Khan, M. CO_2 emissions and sustainable economic development: New evidence on the role of human capital. *Sustain. Dev.* **2020**, *28*, 1279–1288. [CrossRef]
24. Lan, J.; Kakinaka, M.; Huang, X. Foreign Direct Investment, Human Capital and Environmental Pollution in China. *Environ. Resour. Econ.* **2012**, *51*, 255–275. [CrossRef]
25. IInglesi-Lotz, R.; del Corral Morales, L.D. The Effect of Education on a Country's Energy Consumption: Evidence from Developed and Developing Countries (No.678; 2017-33, Issue May). 2017. Available online: https://www.up.ac.za/media/shared/61/WP/wp_2017_33.zp117672.pdf (accessed on 4 April 2022).
26. United Nations Development Programme. *Climate Change Adaptation in Africa: UNDP Synthesis of Experiences and Recommendations*; United Nations Development Programme: New York, NY, USA, 2018. [CrossRef]
27. Grossman, G.; Krueger, A. Environmental impacts of a North American free trade agreement. In *NBER Working Papers Series*; National Bureau of Economic Research: Cambridge, MA, USA, 1991. [CrossRef]
28. Chiu, Y.B. Carbon dioxide, income and energy: Evidence from a non-linear model. *Energy Econ.* **2017**, *61*, 279–288. [CrossRef]
29. Yao, S.; Zhang, S.; Zhang, X. Renewable energy, carbon emission and economic growth: A revised environmental Kuznets curve perspective. *J. Clean. Prod.* **2019**, *235*, 1338–1352. [CrossRef]
30. Ehigiamusoe, K.U.; Lean, H.H.; Babalola, S.J.; Poon, W.C. The roles of financial development and urbanization in degrading environment in Africa: Unravelling non-linear and moderating impacts. *Energy Rep.* **2022**, *8*, 1665–1677. [CrossRef]
31. Mahmood, N.; Wang, Z.; Hassan, S.T. Renewable energy, economic growth, human capital, and CO_2 emission: An empirical analysis. *Environ. Sci. Pollut. Res.* **2019**, *26*, 20619–20630. [CrossRef]
32. Munir, Q.; Lean, H.H.; Smyth, R. CO_2 emissions, energy consumption and economic growth in the ASEAN-5 countries: A cross-sectional dependence approach. *Energy Econ.* **2020**, *85*, 1–10. [CrossRef]

33. Sarkodie, S.A. The invisible hand and EKC hypothesis: What are the drivers of environmental degradation and pollution in Africa? *Environ. Sci. Pollut. Res.* **2018**, *25*, 21993–22022. [CrossRef]
34. Shahbaz, M.; Lean, H.H.; Shabbir, M.S. Environmental Kuznets curve hypothesis in Pakistan: Cointegration and Granger causality. *Renew. Sustain. Energy Rev.* **2012**, *16*, 2947–2953. [CrossRef]
35. Sinha, A.; Shahbaz, M. Estimation of Environmental Kuznets Curve for CO_2 emission: Role of renewable energy generation in India. *Renew. Energy* **2018**, *119*, 703–711. [CrossRef]
36. Tan, F.; Lean, H.H.; Khan, H. Growth and environmental quality in Singapore: Is there any trade-off? *Ecol. Indic.* **2014**, *47*, 149–155. [CrossRef]
37. Wang, Q.; Zhang, F. Does increasing investment in research and development promote economic growth decoupling from carbon emission growth? An empirical analysis of BRICS countries. *J. Clean. Prod.* **2020**, *252*, 119853. [CrossRef]
38. Abid, M. Impact of economic, financial, and institutional factors on CO_2 emissions: Evidence from Sub-Saharan Africa economies. *Util. Policy* **2016**, *41*, 85–94. [CrossRef]
39. Dogan, E.; Ulucak, R.; Kocak, E.; Isik, C. The use of ecological footprint in estimating the Environmental Kuznets Curve hypothesis for BRICST by considering cross-section dependence and heterogeneity. *Sci. Total Environ.* **2020**, *723*, 138063. [CrossRef]
40. Zoundi, Z. CO_2 emissions, renewable energy and the environmental Kuznets curve, a panel cointegration approach. Renew. Sustain. *Energy Rev.* **2017**, *72*, 1067–1075. [CrossRef]
41. Azam, M.; Khan, A.Q.; Zaman, K.; Ahmad, M. Factors determining energy consumption: Evidence from Indonesia, Malaysia and Thailand. *Renew. Sustain. Energy Rev.* **2015**, *42*, 1123–1131. [CrossRef]
42. Ehigiamusoe, K.U.; Guptan, V.; Lean, H.H. Impact of financial structure on environmental quality: Evidence from panel and disaggregated data. *Energy Sources Part B Econ. Plan. Policy* **2020**, *14*, 359–383. [CrossRef]
43. Gu, W.; Zhao, X.; Yan, X.; Wang, C.; Li, Q. Energy technological progress, energy consumption, and CO_2 emissions: Empirical evidence from China. *J. Clean. Prod.* **2019**, *236*, 117666. [CrossRef]
44. Khan, M.K.; Teng, J.Z.; Khan, M.I.; Khan, M.O. Impact of globalization, economic factors and energy consumption on CO_2 emissions in Pakistan. *Sci. Total Environ.* **2019**, *688*, 424–436. [CrossRef]
45. Sarkodie, S.A.; Strezov, V. Effect of foreign direct investments, economic development and energy consumption on greenhouse gas emissions in developing countries. *Sci. Total Environ.* **2019**, *646*, 862–871. [CrossRef] [PubMed]
46. Farhani, S.; Kadria, M.; Guirat, Y. The role of energy and environmental quality in exploring the economic sustainability: A new appraisal in the context of north African countries. *Sustainability* **2021**, *13*, 13990. [CrossRef]
47. Rahman, H.U.; Zaman, U.; Górecki, J. The Role of Energy Consumption, Economic Growth and Globalization in Environmental Degradation: Empirical Evidence from the BRICS Region. *Sustainability* **2021**, *13*, 1924. [CrossRef]
48. Hanif, I.; Aziz, B.; Chaudhry, I.S. Carbon emissions across the spectrum of renewable and nonrenewable energy use in developing economies of Asia. *Renew. Energy* **2019**, *143*, 586–595. [CrossRef]
49. Namahoro, J.P.; Wu, Q.; Zhou, N.; Xue, S. Impact of energy intensity, renewable energy, and economic growth on CO_2 emissions: Evidence from Africa across regions and income levels. *Renew. Sustain. Energy Rev.* **2021**, *147*, 111233. [CrossRef]
50. Sharif, A.; Mishra, S.; Sinha, A.; Jiao, Z.; Shahbaz, M.; Afshan, S. The renewable energy consumption-environmental degradation nexus in Top-10 polluted countries: Fresh insights from quantile-on-quantile regression approach. *Renew. Energy* **2020**, *150*, 670–690. [CrossRef]
51. Bekun, F.V.; Alola, A.A.; Sarkodie, S.A. Toward a sustainable environment: Nexus between CO_2 emissions, resource rent, renewable and nonrenewable energy in 16-EU countries. *Sci. Total Environ.* **2019**, *657*, 1023–1029. [CrossRef]
52. Sun, X.; Zhu, B.K.; Zhang, S.; Zeng, H.; Li, K.; Wang, B.; Dong, Z.; Zhou, C. New indices system for quantifying the nexus between economic-social development, natural resources consumption, and environmental pollution in China during 1978–2018. *Sci. Total Environ.* **2022**, *804*, 150180. [CrossRef] [PubMed]
53. Balsalobre-Lorente, D.; Shahbaz, M.; Roubaud, D.; Farhani, S. How economic growth, renewable electricity and natural resources contribute to CO_2 emissions? *Energy Policy* **2018**, *113*, 356–367. [CrossRef]
54. Shittu, W.; Fatai, F.; Ibrahim, M. An investigation of the nexus between natural resources, environmental performance, energy security and environmental degradation: Evidence from Asia. *Resour. Policy* **2021**, *73*, 102227. [CrossRef]
55. Ahmad, M.; Satrovic, E. Relating fiscal decentralization and financial inclusion to environmental sustainability: Criticality of natural resources. *J. Environ. Manag.* **2023**, *325*, 116633. [CrossRef] [PubMed]
56. Chankrajang, T.; Muttarak, R. Green returns to education: Does schooling contribute to pro-environmental behaviours? Evidence from Thailand. *Ecol. Econ.* **2017**, *131*, 434–448. [CrossRef]
57. Meyer, A. Does education increase pro-environmental behavior? Evidence from Europe. *Ecol. Econ.* **2015**, *116*, 108–121. [CrossRef]
58. Zafar, M.W.; Shahbaz, M.; Sinha, A.; Sengupta, T.; Qin, Q. How renewable energy consumption contribute to environmental quality? The role of education in OECD countries. *J. Clean. Prod.* **2020**, *268*, 122149. [CrossRef]
59. Zafar, M.W.; Sinha, A.; Ahmed, Z.; Qin, Q.; Zaidi, S.A.H. Effects of biomass energy consumption on environmental quality: The role of education and technology in Asia-Pacific Economic Cooperation countries. *Renew. Sustain. Energy Rev.* **2021**, *142*, 110868. [CrossRef]
60. Zafar, M.W.; Qin, Q.; Malik, M.N.; Zaidi, S.A.H. Foreign direct investment and education as determinants of environmental quality: The importance of post Paris Agreement (COP_{21}). *J. Environ. Manag.* **2020**, *270*, 110827. [CrossRef] [PubMed]

61. Maranzano, P.; Bento, J.P.C.; Manera, M. The Role of Education and Income Inequality on Environmental Quality: A Panel Data Analysis of the EKC Hypothesis on OECD Countries. *Sustainability* **2022**, *14*, 1622. [CrossRef]
62. Liu, H.; Alharthi, M.; Atil, A.; Zafar, M.W.; Khan, I. A non-linear analysis of the impacts of natural resources and education on environmental quality: Green energy and its role in the future. *Resour. Policy* **2022**, *79*, 102940. [CrossRef]
63. Zhang, C.; Khan, I.; Dagar, V.; Saeed, A.; Zafar, M.W. Environmental impact of information and communication technology: Unveiling the role of education in developing countries. *Technol. Forecast. Soc. Chang.* **2022**, *178*, 121570. [CrossRef]
64. Boukhelkhal, A. Impact of economic growth, natural resources and trade on ecological footprint: Do education and longevity promote sustainable development in Algeria? *Int. J. Sustain. Dev. World Ecol.* **2022**, *29*, 875–887. [CrossRef]
65. Katircioglu, S.; Katircioĝlu, S.; Saqib, N. Does higher education system moderate energy consumption and climate change nexus? Evidence from a small island. *Air Qual. Atmos. Health* **2020**, *13*, 153–160. [CrossRef]
66. Osuntuyi, B.V.; Lean, H.H. Economic growth, energy consumption and environmental degradation nexus in heterogeneous countries: Does education matter? *Environ. Sci. Eur.* **2022**, *34*, 2–16. [CrossRef]
67. Osuntuyi, B.V.; Lean, H.H. Environmental degradation, Economic Growth, and Energy Consumption: The Moderating Role of Education. *Sustain. Dev.* **2023**, 1–12. [CrossRef]
68. Ahmed, Z.; Zafar, M.W.; Ali, S.; Danish. Linking urbanization, human capital, and the ecological footprint in G7 countries: An empirical analysis. *Sustain. Cities Soc.* **2020**, *55*, 102064. [CrossRef]
69. Ahmed, Z.; Wang, Z. Investigating the impact of human capital on the ecological footprint in India: An empirical analysis. Environ. *Sci. Pollut. Res.* **2019**, *26*, 26782–26796. [CrossRef]
70. Pata, U.K.; Caglar, A.E. Investigating the EKC hypothesis with renewable energy consumption, human capital, globalization and trade openness for China: Evidence from augmented ARDL approach with a structural break. *Energy* **2021**, *216*, 119220. [CrossRef]
71. Sapkota, P.; Bastola, U. Foreign direct investment, income, and environmental pollution in developing countries: Panel data analysis of Latin America. *Energy Econ.* **2017**, *64*, 206–212. [CrossRef]
72. Bano, S.; Zhao, Y.; Ahmad, A.; Wang, S.; Liu, Y. Identifying the impacts of human capital on carbon emissions in Pakistan. *J. Clean. Prod.* **2018**, *183*, 1082–1092. [CrossRef]
73. Ahmed, Z.; Nathaniel, S.P.; Shahbaz, M. The criticality of information and communication technology and human capital in environmental sustainability: Evidence from Latin American and Caribbean countries. *J. Clean. Prod.* **2021**, *286*, 125529. [CrossRef]
74. Tang, C.F.; Abosedra, S.; Naghavi, N. Does the quality of institutions and education strengthen the quality of the environment? Evidence from a global perspective. *Energy* **2021**, *218*, 119303. [CrossRef]
75. World Bank. World Development Indicators | DataBank 2021. Available online: https://databank.worldbank.org/source/world-development-indicators (accessed on 10 July 2021).
76. Global Footprint Network. Ecological Footprint. *Encycl. Ecol.* **2008**, 1031–1037. [CrossRef]
77. EIA. U.S. Energy Information Administration. 2021. Available online: https://www.eia.gov/international/data/world (accessed on 10 July 2021).
78. Breusch, T.S.; Pagan, A.R. The lagrange multiplier test and its applications to model specification in Econometrics. *Rev. Econ. Stud.* **1980**, *47*, 239–252. [CrossRef]
79. Baltagi, B.H.; Feng, Q.; Kao, C. A lagrange multiplier test for cross-sectional dependence in a fixed effects panel data model. *J. Econom.* **2012**, *170*, 164–177. [CrossRef]
80. Pesaran, M.H. *General Diagnostic Tests for Cross-Section Dependence in Panels*; No. 1240; IZA Discussion Paper Series; Institute for the Study of Labor: Bonn, Germany, 2004. [CrossRef]
81. Pesaran, M.H. A simple panel unit root test in the presence of cross-section dependence. *J. Appl. Econom.* **2007**, *22*, 265–312. [CrossRef]
82. Westerlund, J. Testing for error correction in panel data. *Oxf. Bull. Econ. Stat.* **2007**, *69*, 709–748. [CrossRef]
83. Gujarati, D.N.; Porter, D.C. *Basic Econometrics*, 5th ed.; McGraw-Hill International Edition: New York, NY, USA, 2009.
84. Driscoll, J.C.; Kraay, A.C. Consistent covariance matrix estimation with spatially dependent panel data. *Rev. Econ. Stat.* **1998**, *80*, 549–559. [CrossRef]
85. Hoechle, D. Robust standard errors for panel regressions with cross-sectional dependence. *Stata J.* **1995**, *7*, 281–312. [CrossRef]
86. He, Z.; Xu, S.; Shen, W.; Long, R.; Chen, H. Impact of urbanization on energy related CO_2 emission at different development levels: Regional difference in China based on panel estimation. *J. Clean. Prod.* **2017**, *140*, 1719–1730. [CrossRef]
87. Özokcu, S.; Özdemir, Ö. Economic growth, energy, and environmental Kuznets curve. *Renew. Sustain. Energy Rev.* **2017**, *72*, 639–647. [CrossRef]
88. Danish Baloch, M.A.; Wang, B. Analyzing the role of governance in CO_2 emissions mitigation: The BRICS experience. *Struct. Chang. Econ. Dyn.* **2019**, *51*, 119–125. [CrossRef]
89. Le, T.; Chang, Y.; Park, D. Governance, environmental vulnerability, and PM2.5 concentrations: International evidence. *Energy J.* **2021**, *42*, 1–24. [CrossRef]
90. Pedroni, P. Fully modified OLS for heterogeneous cointegrated panels. In *Nonstationary Panels, Panel Cointegration, and Dynamic Panels*; Emerald Group Publishing Limited: Bingley, UK, 2000; Volume 15, pp. 93–130. Available online: https://pdfs.semanticscholar.org/b6b4/fe66e3344b173e4cd91c9ec768296c2e4fbf.pdf (accessed on 9 November 2022).

91. Bayale, N.; Ali, E.; Tchagnao, A.F.; Nakumuryango, A. Determinants of renewable energy production in WAEMU countries: New empirical insights and policy implications. *Int. J. Green Energy* **2021**, *18*, 602–614. [CrossRef]
92. Danish; Ulucak, R.; Khan, S.U.D. Determinants of the ecological footprint: Role of renewable energy, natural resources, and urbanization. *Sustain. Cities Soc.* **2020**, *54*, 101996. [CrossRef]
93. Shahbaz, M.; Sarwar, S.; Chen, W.; Malik, M.N. Dynamics of electricity consumption, oil price and economic growth: Global perspective. *Energy Policy* **2017**, *108*, 256–270. [CrossRef]
94. Rahman, M.M. Do population density, economic growth, energy use and exports adversely affect environmental quality in Asian populous countries? *Renew. Sustain. Energy Rev.* **2017**, *77*, 506–514. [CrossRef]
95. Raza, S.A.; Shah, N. Testing environmental Kuznets curve hypothesis in G7 countries: The role of renewable energy consumption and trade. *Environ. Sci. Pollut. Res.* **2018**, *25*, 26965–26977. [CrossRef]
96. Kao, C.; Chiang, M.H. On the estimation and inference of a cointegrated regression in panel data. In *Nonstationary Panels, Cointegration in Panels and Dynamic Panels*; Elsevier Science Inc.: Amsterdam, The Netherlands, 2000; Volume 15, pp. 179–222. [CrossRef]
97. Maji, I.K.; Sulaiman, C.; Abdul-Rahim, A.S. Renewable energy consumption and economic growth nexus: A fresh evidence from West Africa. *Energy Rep.* **2019**, *5*, 384–392. [CrossRef]
98. Dogan, E.; Seker, F.; Bulbul, S. Investigating the impacts of energy consumption, real GDP, tourism and trade on CO_2 emissions by accounting for cross-sectional dependence: A panel study of OECD countries. *Curr. Issues Tour.* **2017**, *20*, 1701–1719. [CrossRef]
99. Brambor, T.; Clark, W.R.; Golder, M. Understanding interaction models: Improving empirical analyses. *Polit. Anal.* **2006**, *14*, 63–82. [CrossRef]
100. International Energy Agency. Africa Energy Outlook 2019. In *World Energy Outlook Report*; IEA: Paris, France, 2019; pp. 1–285. Available online: https://www.iea.org/reports/africa-energy-outlook-2019 (accessed on 26 April 2020).
101. Balsalobre-Lorente, D.; Sinha, A.; Driha, O.M.; Mubarik, M.S. Assessing the impacts of ageing and natural resource extraction on carbon emissions: A proposed policy framework for European economies. *J. Clean. Prod.* **2021**, *296*, 126470. [CrossRef]
102. Nathaniel, S.P.; Nwulu, N.; Bekun, F. Natural resource, globalization, urbanization, human capital, and environmental degradation in Latin American and Caribbean countries. *Environ. Sci. Pollut. Res.* **2021**, *28*, 6207–6221. [CrossRef]
103. Zafar, M.W.; Saleem, M.M.; Destek, M.A.; Caglar, A.E. The dynamic linkage between remittances, export diversification, education, renewable energy consumption, economic growth, and CO_2 emissions in top remittance-receiving countries. *Sustain. Dev.* **2021**, *30*, 165–175. [CrossRef]
104. Ohlan, R. The impact of population density, energy consumption, economic growth and trade openness on CO_2 emissions in India. *Nat. Hazards* **2015**, *79*, 1409–1428. [CrossRef]
105. Wang, Y.; Kang, Y.; Wang, J.; Xu, L. Panel estimation for the impacts of population-related factors on CO2 emissions: A regional analysis in China. *Ecol. Indic.* **2017**, *78*, 322–330. [CrossRef]
106. United Nations Environment Programme. Africa Environment Tracking: Issues and Developments. UN: New York, NY, USA, 2005; pp. 1–50. Available online: https://wedocs.unep.org/bitstream/handle/20.500.11822/8697/-AfricaEnvironmentTracking_IssuesandDevelopments-2005428.pdf?sequence=3&isAllowed= (accessed on 11 July 2020).
107. United Nations. Population: Our Growing Population. UN: New York, NY, USA, 2020; pp. 1–5. Available online: https://www.un.org/en/sections/issues-depth/population/ (accessed on 2 October 2020).
108. Destek, M.A.; Sinha, A. Renewable, non-renewable energy consumption, economic growth, trade openness and ecological footprint: Evidence from organisation for economic Co-operation and development countries. *J. Clean. Prod.* **2020**, *242*, 118537. [CrossRef]

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.